

WIPRO TALENTNEXT PROGRAM

PBL HANDS-ON ASSIGNMENT

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Date: 04-07-2026

Topic: Normalization

Question 1

Observe the structure of the given EMPLOYEE table below and suggest if this table can be further normalized.

EMPLOYEE (EMPNO, ENAME, SAL, DEPTNO, DNAME, LOC)

Explain what normal form this table is in and how you can make this into the next normal form.

Answer:

The given table is in First Normal Form (1NF) because:

All attributes contain atomic values.

There are no repeating groups.

EMPLOYEE(EMPNO, ENAME, SAL, DEPTNO, DNAME, LOC)

In these Table There is a transitive dependency.

So DNAME and LOC do not depend directly on EMPNO.

Split the table into two tables.

For Example

EMPNO	ENAME	SAL	DEPTNO	DNAME	LOC
101	Ravi	50000	D01	HR	Hyderabad
102	Priya	45000	D02	Finance	Chennai

Normalized Table:

EMPLOYEE

EMPNO	ENAME	SAL	DEPTNO
101	Ravi	50000	D01
102	Priya	45000	D02

DEPARTMENT

DEPTNO	DNAME	LOC
D01	HR	Hyderabad
D02	Finance	Chennai

Question 2

Observe the structure of the given STUDENT table and suggest if this table can be further normalized.

STUDENT (ROLLNO, NAME, AGE, EXAM, MARKS, GRADE)

Explain what normal form this table is in and how you can make this into the next normal form.

Answer

The given STUDENT table is in First Normal Form because:

All attributes contain atomic values.

There are no repeating groups.

Example Table

ROLLNO	NAME	AGE	EXAM	MARKS	GRADE
1	Ravi	20	DBMS	92	A
2	Priya	21	DBMS	78	B

In these Table There is a transitive dependency.

GRADE depends on MARKS, not directly on ROLLNO.

Split the table into the following tables.

STUDENT

ROLLNO	NAME	AGE
1	Ravi	20
2	Priya	21

RESULT

ROLLNO	EXAM	MARKS
1	DBMS	92
2	DBMS	78

GRADE

MARKS RANGE	GRADE
90-100	A
75-89	B
60-74	C

Question 3

Observe the structure of a given EMPLOYEE table and suggest what kind of problem this table has and how to solve the problem EMPNO, PROJECT_NO, NO_OF_DAYS, CUSTOMERNAME In the above table EMPNO AND PROJECT_NO both are together a composite primary key.

Answer:

The table has Partial Dependency.

CUSTOMERNAME depends only on PROJECT_NO

EMPNO, PROJECT_NO are composite primary key.

Spilt the Table into two tables

1. EMP (EMPNO, PROJECT_NO, NO_OF_DAYS)
2. PROJECT (PROJECT_NO, CUSTOMERNAME)

This removes the partial dependency and brings the table into Second Normal Form.

For example:

Original Table:

EMPNO	PROJECT_NO	NO_OF_DAYS	CUSTOMERNAME
1	P101	20	A
2	P102	15	B

Split into 2 tables.

EMP

EMPNO	PROJECT_NO	NO_OF_DAYS
1	P101	20
2	P102	15

PROJECT

PROJECT_NO	CUSTOMERNAME
P101	A
P102	B